

Study on Electrode Potential of Capacitance Proximity Fuze

Zhang Xin, Cui Zhanzhong, Xu Lixin

National Defense Key Laboratory of Mechatronics Engineering and Control,
Beijing Institute of Technology, Beijing

Email: cocohawkxin@163.com

Abstract –Detection mechanism of capacitance proximity fuze is the base of fuze design work. Electric field and circuit methods are used to calculate the potential proportional relationship between the two electrodes. The potential of electrode I is obtained through experiment. Thus we can get the potential of electrode II. Pispce software is used to simulate the whole circuit, and get the simulation result of potential of electrode II. The experiment comes to the conclusions as following: the potential of electrode II decreases when fuze gets close to the target; and the potential proportional relationship between the two electrodes based on the two methods are nearly same; potential of electrode II based on theory and experiment fits the result of simulation.

Keywords –detection mechanism of capacitance proximity fuze, electrode potential, potential proportion, Pispce.

I. INTRODUCTION

Currently, there are two types of detectors used in the capacitance proximity fuze. One is frequency discriminated detector, and the other is direct-coupled detector[1]. While, according to the number of electrodes, the capacitance proximity fuze detector is divided into two types: two electrodes and three electrodes. The paper focus on the detector of the two electrodes that is direct-coupled.

Direct-coupled detector is a type of amplitude discriminated detector, which means that the amplitude or variation of output signal of detect electrode (electrode II) is measured to detect target.

In this paper, the potential proportional relationship between the two electrodes is detected through circuit and electric field methods. Then get the potential of electrode I through experiment of the electrodes getting close to target. At last, Pispce software is used to simulate the whole circuit, and get the output signal of detection circuit module, which is to detect the output of detect electrode(electrode II).

II. POTENTIAL PROPORTION CALCULATION

The two electrodes type detector consists of oscillator, electrodes and detection circuit. The electrode which is connected with the oscillator is called electrode I, while the other one connected with the detection circuit is called electrode II. The schematic of the detector is shown in Figure 1.

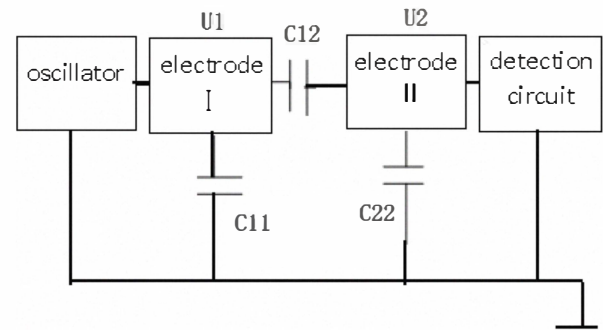


Fig 1 The schematic of the detector

A. Potential proportion caculation with circuit method

Here, the ground is considered as the target. Based on the circuit method, the mutual capacitance between electrode I and electrode II is called c_{12} , and their self capacitance are called c_{11} 、 c_{22} , which are the capacitance between the electrode and the ground. The electrode I is the output of oscillator, and the electrode II is as the input of the detection circuit. So there is a electric field between the two electrodes, and the region closed to the electrodes is taken as quasi-static electric field [2]. U_1 is potential of electrode I, and U_2 is the potential of electrode II. c_{12} and c_{22} constitute a series branch. Based on the voltage theorem in series branch, the relation of U_1 and U_2 is as below:

$$U_2 = \frac{c_{12}}{c_{12} + c_{22}} U_1 \quad (1)$$

Here $K_1 = \frac{c_{12}}{c_{12} + c_{22}}$. When the fuze is approaching to

the target, the capacitance is changing, so K_1 and U_2 are changing.

B. Potential proportion calculation with electric field method

For multi-conductor system, the relation of the charge and the potential is :

$$\left. \begin{aligned} q_1 &= \beta_{11}u_1 + \beta_{12}u_2 + \cdots + \beta_{1n}u_n \\ q_2 &= \beta_{21}u_1 + \beta_{22}u_2 + \cdots + \beta_{2n}u_n \\ &\vdots \\ q_n &= \beta_{n1}u_1 + \beta_{n2}u_2 + \cdots + \beta_{nn}u_n \end{aligned} \right\} \quad (2)$$

β_{ij} is capacitance coefficient[3-4],and the relation between capacitance and capacitance coefficient is :

$$C_{ii} = \sum_{j=1}^n \beta_{ij} \quad (3)$$

$$C_{ij} = -\beta_{ij} \quad (4)$$

So get

$$\begin{cases} q_1 = \beta_{11}u_1 + \beta_{12}u_2 \\ q_2 = \beta_{21}u_1 + \beta_{22}u_2 \end{cases} \quad (5)$$

$$\beta_{11} = c_{11} + c_{12}$$

$$\beta_{12} = -c_{12}$$

$$\beta_{21} = c_{21} + c_{22}$$

$$\beta_{22} = -c_{22} \quad (6)$$

According to (5),get

$$\begin{cases} u_1 = p_{11}q_1 + p_{12}q_2 \\ u_2 = p_{21}q_1 + p_{22}q_2 \end{cases} \quad (7)$$

Based on (5) and (7),get

$$p = [\beta]^{-1} \quad (8)$$

Together with (6),get

$$p_{21} = \frac{-c_{21}}{(c_{11} + c_{12})(c_{21} + c_{22}) - c_{12}^2} \quad (9)$$

$$p_{22} = \frac{c_{21} + c_{22}}{(c_{11} + c_{12})(c_{21} + c_{22}) - c_{12}^2} \quad (10)$$

At last,get

$$\begin{aligned} u_2 = & \frac{c_{21} + c_{22}}{(c_{11} + c_{12})(c_{21} + c_{22}) - c_{12}^2} \\ & * ((c_{11} + c_{12})u_1 + (-c_{12})u_2) \\ & + \frac{c_{21} + c_{22}}{(c_{11} + c_{12})(c_{21} + c_{22}) - c_{12}^2} \\ & * ((-c_{21})u_1 + (c_{21} + c_{22})u_2) \end{aligned} \quad (11)$$

So

$$u_2 = \frac{-c_{21}(c_{11} + c_{12}) - c_{21}(c_{21} + c_{22})}{(c_{11} + c_{12})(c_{21} + c_{22}) - 3c_{12}^2 - (c_{21} + c_{22})^2} u_1 \quad (12)$$

Now get

$$K_2 = \frac{-c_{21}(c_{11} + c_{12}) - c_{21}(c_{21} + c_{22})}{(c_{11} + c_{12})(c_{21} + c_{22}) - 3c_{12}^2 - (c_{21} + c_{22})^2} \quad (13)$$

C. Results comparison

Now, K_1 and K_2 are obtained, and they are the functions of capacitance. So the capacitance should be obtained to compare K_1 and K_2 .

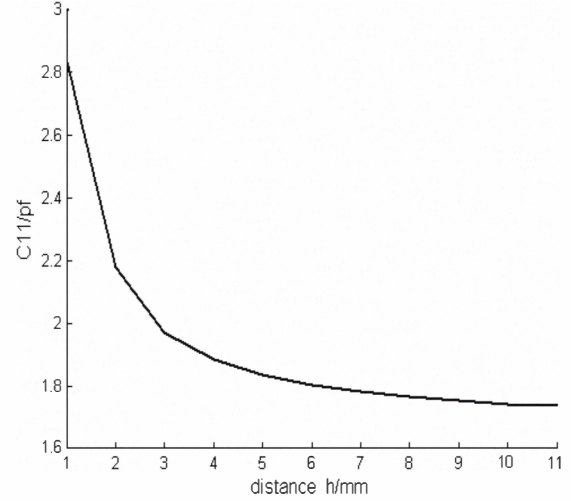
For capacitance fuze, the capacitance is affected with the size of the electrode, the position of the electrode, the distance of the electrode and the angle between the fuze and the target[5].From reference[6],we get the capacitance function:

$$c_{11}(h) = 2.3 - 3 \cdot 10^{-3}h + 6.3 \cdot 10^{-6} \cdot h^3 \quad (14)$$

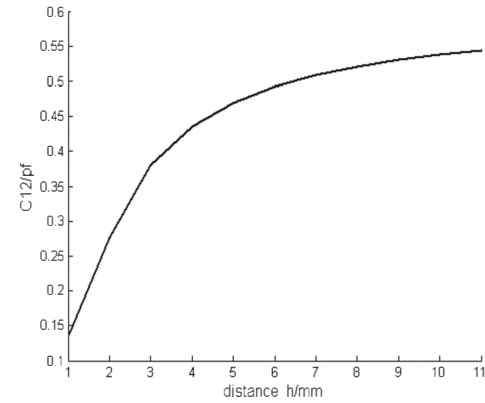
$$c_{22}(h) = 9.9 - 4.7 \cdot 10^{-3}h + 9 \cdot 10^{-6} \cdot h^3 \quad (15)$$

$$c_{12}(h) = 0.21 + 1.42 \cdot 10^{-3}h - 2.6 \cdot 10^{-6} \cdot h^3 \quad (16)$$

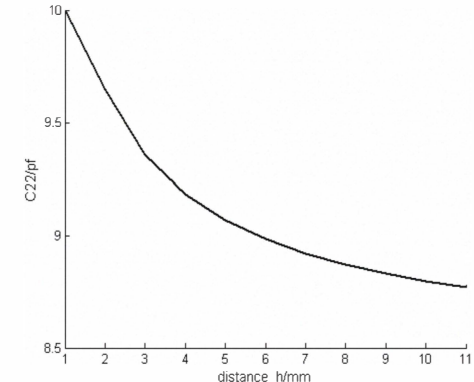
h is the distance between the fuze and target. C_{11} 、 C_{12} and C_{22} are shown as Figure 2.



(a) The curve of C_{11}

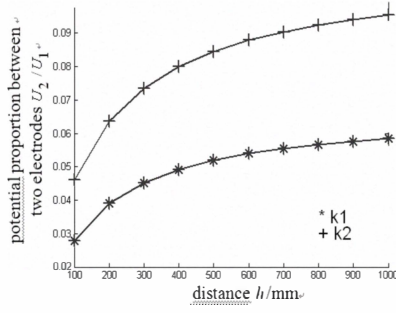


(b) The curve of C_{12}



(c) The curve of C_{22}

Based on the capacitance function, $K_1(h)$ and $K_2(h)$ can be get by MATLAB, and the curve is shown in Figure 3.

Fig 3 The curve of K_1 and K_2

From Fig 2, the trend of K_1 and K_2 are the same, and it shows that U_2 decreases when fuze gets close to the target.

III. POTENTIAL OF ELECTROD I

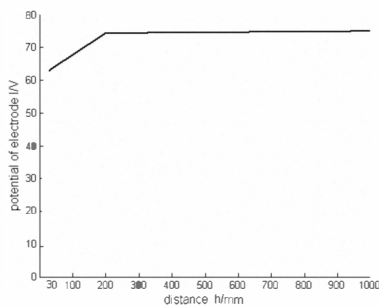
In the analysis above, potential of electrode I is represented by U_1 , which is the output of oscillation circuit. Here, shell circuit is used as oscillator, the equivalent capacitance between electrode I and electrode II is as the small capacitance in parallel with inductor. When fuze (electrodes) gets close to target, the equivalent capacitance between electrode I and electrode II would change, which would lead to the change of the frequency and the amplitude of oscillator output signal (electrode I). For the coupled detector, we only care about the amplitude of the signal.

Together with the analysis of potential proportional relationship between the two electrodes above, we know that the potential change of electrode I will affect the potential of electrode II. Now, to get the potential of electrode I, an experiment is designed to simulate the process of fuze (electrodes) getting close to target.

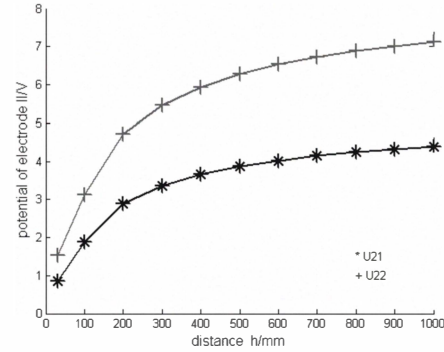
As frequency of oscillator is high, and the oscilloscope memory space is limited, we can't record the whole data of the process. So divide the distance into 10 intervals, and record the data at some nodes. Do the same experiment 5 times, and average 5 experimental data, then fit the experimental average data as:

$$U_1 = \begin{cases} 74.29 + 0.0007625 \times h & 1000 > h > 200 \\ 63.01 + 0.0664 \times h & 200 > h > 30 \end{cases} \quad (17)$$

$U_1(h)$ is shown as Figure 4.

Fig 4 The curve of $U_1(h)$

Now, the potential proportional relationship between the two electrodes $K_1(h)$, $K_2(h)$, and the potential of electrode I U_1 are obtained. So the potential of electrode II $U_{21}(h)$ (based on $K_1(h)$) and $U_{22}(h)$ (based on $K_2(h)$) are shown as Figure 5.

Fig 5 The curve of $U_{21}(h)$ and $U_{22}(h)$

IV. CIRCUIT SIMULATION

The Pspice software is used to simulate the whole circuit. Based on the formula (20) - (22), the variation of C_{12} is much less than the ones of C_{11} and C_{22} , so a constant capacitance is represented as C_{12} . Varactor diodes are represented as C_{11} and C_{22} , which are controlled by customized source [7]. At last, the output signal of electrode II is processed by detection circuit module. The part of simulation circuit is shown as Figure 6.

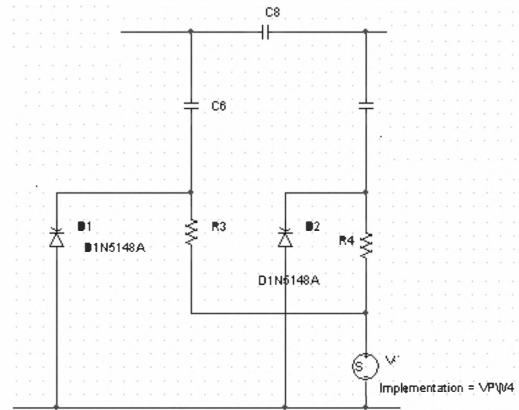
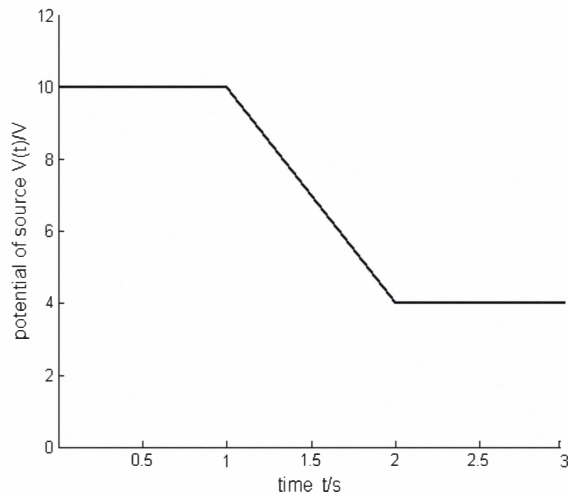


Fig 6 The schematic of part simulate circuit

For simulation circuit, parameters mutation can impact on final output. To avoid the effect, set the source as below:

$$V(t) = \begin{cases} 10 & 0 \leq t < 1 \\ 10 - 6(t-1) & 1 \leq t < 2 \\ 4 & 2 \leq t \leq 3 \end{cases} \quad (18)$$

$V(t)$ is shown as Figure 7.

Fig 7 The curve of the source $V(t)$

According to the formula, select the appropriate diode, and calculate the range of controlled sources. Simulation results of the output of the whole circuit is shown as Figure 8.

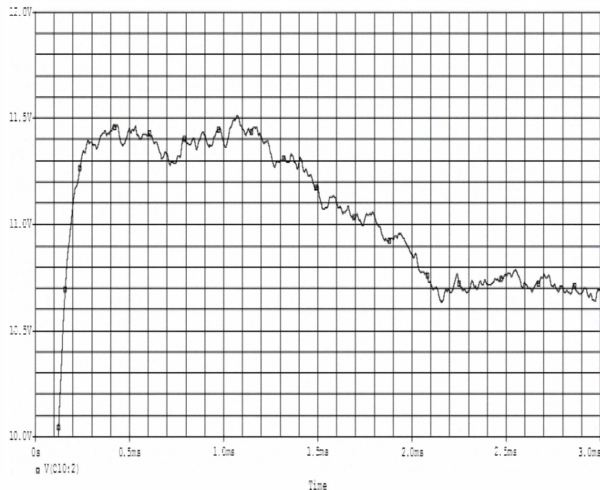


Fig 8 The output curve of the whole circuit

CONCLUSION

The circuit and electric field methods are used to get the potential proportional relationship between the two electrodes, obtain the potential of electrode I through experiment, and get the potential of electrode II. Then Pispce software is used to simulate the whole circuit, and get the simulation result of potential of electrode II. The conclusions of the experiment is as following: the potential of electrode II decreases when fuze gets close to the target; and the potential proportional relationship between the two electrodes based on the two method are nearly same; potential of electrode II based on theory and experiment fits the result of simulation.

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AUTHOR BIOGRAPHY

Zhang Xin was born in Shijiazhuang, China, in 1984. She is a PhD candidate in Beijing Institute of Technology, China. Her research interests include capacitance proximity fuze.